1.Negative or zero

```python
n = float(input("Enter a number: "))

if n > 0:
    print("Positive")
elif n < 0:
    print("Negative")
else:
    print("Zero")
```

Input: Enter a number: 5

Output: Positive

2.Even or odd

```python
n = int(input("Enter a number: "))

if n % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Input: Enter a number: 8

Output: Even

3.Leap year

```python
year = int(input("Enter year: "))

if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

Input: Enter year: 2024

Output: Leap Year

4.MAX

```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a > b:
    print("Maximum =", a)
else:
    print("Maximum =", b)
```

Input: 10, 20

Output: Maximum = 20

5.MIN

```python
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a < b:
    print("Minimum =", a)
else:
    print("Minimum =", b)
```

Input: 10, 20

Output: Minimum = 10

6.Palindrome

```
n = input("Enter a number: ")

if n == n[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Input: Enter a number: 121

Output: Palindrome

7.Prime or not

```
n = int(input("Enter a number: "))

if n > 1:
    for i in range(2, n):
        if n % i == 0:
            print("Not Prime")
            break
    else:
        print("Prime")
else:
    print("Not Prime")
```

Input: Enter a number: 13

Output: Prime

8.Divisible by 3 or 5

```
n = int(input("Enter a number: "))

if n % 3 == 0 or n % 5 == 0:
    print("Divisible")
else:
    print("Not Divisible")
```

Input: Enter a number: 15

Output: Divisible

9.vow or cons

```
ch = input("Enter a letter: ")

if ch.lower() in "aeiou":
    print("Vowel")
else:
    print("Consonant")
```

Input: Enter a letter: a

Output: Vowel

10.Valid

```
n = int(input("Enter a number: "))

if 0 <= n <= 100:
    print("Valid")
else:
    print("Invalid")
```

Input: Enter a number: 75

Output: Valid

11.Sum Even

```
n = int(input("Enter limit: "))
sum_even = 0

for i in range(2, n + 1, 2):
    sum_even += i

print("Sum =", sum_even)
```

Input: Enter limit: 10

Output: Sum = 30

12.Sum odd

```
n = int(input("Enter limit: "))
sum_odd = 0

for i in range(1, n + 1, 2):
    sum_odd += i

print("Sum =", sum_odd)
```

Input: Enter limit: 10

Output: Sum = 25

13.Average

```
n = int(input("How many numbers? "))
total = 0

for i in range(n):
    num = int(input("Enter number: "))
    total += num

print("Average =", total / n)
```

Input: 3 numbers: 10,20,30

Output: Average = 20.0

14.SQRT

```
import math

n = float(input("Enter a number: "))
print("Square Root =", math.sqrt(n))
```

Input: Enter a number: 25

Output: Square Root = 5.0

15.MAX prime

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

max_prime = -1

for n in range(a, b + 1):
    if n > 1:
```

```
        for i in range(2, n):
            if n % i == 0:
                break
        else:
            max_prime = n

print("Maximum Prime =", max_prime)
```

Input: 10, 20

Output: Maximum Prime = 19

16.Quadratic equation

```
import math

a = float(input("Enter a: "))
b = float(input("Enter b: "))
c = float(input("Enter c: "))

d = b*b - 4*a*c

if d >= 0:
    r1 = (-b + math.sqrt(d)) / (2*a)
    r2 = (-b - math.sqrt(d)) / (2*a)
    print("Roots:", r1, r2)
else:
    print("No Real Roots")
```

Input: a=1, b=-3, c=2

Output: Roots: 2.0 1.0

17.GCD of 3 num

```
import math

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

gcd = math.gcd(math.gcd(a, b), c)
print("GCD =", gcd)
```

Input: 12, 18, 24

Output: GCD = 6

18.Valid password

```
password = input("Enter password: ")

if len(password) >= 8:
    print("Valid Password")
else:
    print("Invalid Password")
```

Input: Enter password: abc12345

Output: Valid Password

19.Area of circle

```
radius = float(input("Enter radius: "))
area = 3.14 * radius * radius
```

```
print("Area =", area)
```

Input: Enter radius: 5

Output: Area = 78.5